

Jonah Elkhouri

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Portfolio: <https://jelkhouri.github.io/projects/>
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EDUCATION

University of Virginia, School of Engineering and Applied Science

Charlottesville, VA

B.S. Electrical Engineering (3.83/4.0)

Expected Graduation: May 2028

Relevant Coursework: Probability, Statistics, Introduction to Embedded Computer Systems, Applied Circuits

WORK EXPERIENCE

University of Virginia, Teaching Assistant (Applied Probability)

Fall 2026

- Explained mathematical concepts independently to students
- Graded coding projects and homework assignments

Abernethy & Spencer Garden Center, Operations Intern

Summer 2026

- Implemented Claude to help assess inventory in a recent acquisition over 17,000 unique SKUs
- Developed Python scripts to clean messy inventory data, automating 5000+ edits and saving hours
- Assisted in various construction and restoration projects around the property

Hepburn & Sons LLC, Engineering Intern

Summer 2024

- Conducted research on U.S. Navy SBIR \$1m+ solicitations and emerging energy technologies
- Analyzed current DoD modernization initiatives, such as shipboard DC power distribution
- Researched modular transformer units and electromagnetic aircraft launch systems

Chick-fil-A, Personnel Trainer

June 2023 – August 2025

- Trained and onboarded 15+ new employees, and monitored general performance
- Regularly conducted performance evaluations and related results to management

Desktop Computer Building, Freelance

Summer 2023

- Assisted 15+ clients with the assembly and setup of Windows gaming desktops
- Created parts lists in keeping with budgets from \$500 – \$1000
- Demonstrated and explained technical concepts to middle school students

LEADERSHIP AND INVOLVEMENT

Theme Park Engineering Group, Competitions Chair, Executive

Fall 2025 – Present

- Leading the full design and build of a working model rollercoaster for an intercollegiate competition
- Reviewed resumes to select qualified leaders for sub-teams
- Working in KiCad to design a custom electrical system and control panel
- Integrated positional sensors, lift motors, and brake actuators into the layout
- Using NoLimits2, SolidWorks, and FEA analysis to design custom track and supports

Virginia Undergraduate Law Review, Editor

Fall 2025

- Worked with a writer to produce professional legal analysis for a selective publication
- Provided analytical feedback and review of argumentative and explanatory pieces

PROJECTS

Variable Boost Converter, Applied Circuits

Spring 2026

- Collaborated with a team to design, manufacture, and test a 5V – 12V DC boost converter
- Analytically verified component values, created PCB layouts, and tested the board

Electoral Futures Markets Analysis, Independent

Summer 2026 – Present

- Automated calls to Kalshi and Google Trends APIs using Python scripts
- Analyzed correlations between search trends and electoral betting market prices
- Implemented various libraries (e.g., matplotlib and NumPy) to create useful visualizations

Counterweight Trebuchet, Lead CAD Designer

Spring 2025

- Led the CAD design of a medieval siege weapon to deliver 10 lb. projectiles 270 ft.
- Created materials lists and cost estimates at various stages of the project
- Took part in the testing and assembly of the trebuchet

SKILLS AND TOOLS

Software: VSCode, Claude Code, Arduino IDE, KiCad Circuit Design, Fusion360, SolidWorks, NoLimits2

Programming: Python (matplotlib, NumPy, pandas), Java, C, Git Repo

Technical: Circuit Testing and Analysis, Quantitative Analysis, Troubleshooting, Monte Carlo Simulation

Hardware: Arduino Uno, ESP32, LoRa Radio, AnalogDiscovery2, 3D Printing, Desktop Building, Wiring, Soldering